DiskANN: Algorithm and Theoretical Guarantees

Wook-Shin Han

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Problem: $(1+\varepsilon)$-Approximate Nearest Neighbor

We study a metric space $(\mathcal{M}, D)$ with finite dataset $P \subseteq \mathcal{M}$ and queries $q \in \mathcal{M}$.

Definition 1 (Nearest Neighbor)

Given finite $P \subseteq \mathcal{M}$ and query $q \in \mathcal{M}$:

$$p^* = \arg\min_{p \in P} D(p, q), \quad p^* \in P.$$

Definition 2 ($(1+\varepsilon)$-ANN)

For $\varepsilon > 0$, a point $p \in P$ is an $(1+\varepsilon)$-ANN of $q \in \mathcal{M}$ if

$$D(p, q) \leq (1+\varepsilon)\, D(p^*, q).$$

Motivation & Key Idea

Motivation. Exact NN is expensive at scale; proximity graphs trade off speed/accuracy.

Definition 3 ($(1+\varepsilon)$-Proximity Graph (PG))

Let $G = (P, E)$ be a directed graph on $P \subseteq \mathcal{M}$. G is an $(1+\varepsilon)$-PG if from any start $p_0 \in P$, a greedy/beam walk returns an $(1+\varepsilon)$-ANN in P for any $q \in \mathcal{M}$.

Key idea. Make local moves $p \mapsto u \in N^+(p)$ that monotonically reduce $D(\cdot, q)$ and terminate at an $(1+\varepsilon)$-ANN.

Beam Search (Pseudocode)

```
beam = [start]
for t in 1..max_steps:
    candidates = []
    for s in beam:
        candidates += expand(s)
    if not candidates: break
    beam = top_B(candidates, key=f, B)
    if goal in beam: return best_goal(beam)
```

k-NN Beam Search on a Graph

```
def beam_knn(G, q, k, B):
    Beam = MinMaxHeap()            # frontier over P
    Visited = set()
    start = entrypoint(G)
    Beam.push((D(start, q), start))
    R = MaxHeap(capacity=k)

    while Beam and not all_visited(Beam, Visited):
        _, p = Beam.pop_min()
        if p in Visited: continue
        Visited.add(p)

        d = D(p, q)
        if len(R) < k: R.push((d,p))
        elif d < R.max_key(): R.replace_max((d,p))

        for u in out_neighbors(G, p):
            if u not in Visited:
                Beam.push((D(u,q), u))
                if len(Beam) > B: Beam.pop_max()

    return [pt for (_, pt) in R.items_sorted()]
```

Out-Edge Selection at Node p (α-shortcut greedy)

Input: $P \setminus \{p\}$, parameter $\alpha > 1$.
Procedure:

1. $L \leftarrow$ the list of points of $P \setminus \{p\}$ sorted in ascending order of $D(p, \cdot)$
2. **while** L not empty **do**
   1. $u \leftarrow$ the next point in L
   2. create an edge from p to u (i.e., add u to $N^+(p)$)
   3. remove from L *every* point v satisfying

$$D(u, v) < \frac{1}{\alpha} D(p, v)$$

(note: u itself is also removed)

Shortcut Property (definition and proof)

Definition 4 (Shortcut Property at p with parameter $\alpha > 1$)

For every $v \in P \setminus \{p\}$,

$$v \in N^+(p) \quad \text{or} \quad \exists\, u \in N^+(p) : \; D(u, v) < \frac{1}{\alpha} D(p, v).$$

Lemma 5 (Out-edge selection $\Rightarrow$ Shortcut)

The greedy construction at p ensures the shortcut property holds at p for every $v \in P \setminus \{p\}$.

Proof.

Process $L = P \setminus \{p\}$ in nondecreasing order of $D(p, \cdot)$. Fix $v \in L$. If v is kept, then $v \in N^+(p)$ and we are done. If v is skipped, the rule guarantees the existence of a witness $u \in S \subseteq N^+(p)$ such that $D(u, v) \leq \frac{1}{\alpha} D(p, v)$. Because witnesses are never removed, this inequality remains valid in the final $N^+(p)$. Hence, for every v, either $v \in N^+(p)$ or there exists $u \in N^+(p)$ with the stated inequality; i.e., the shortcut property holds. $\square$

Fundamental Inequalities (when p is not an $(1+\varepsilon)$-ANN)

Let $p^* = \mathrm{NN}(q) \in P$. Assume p is not an $(1+\varepsilon)$-ANN for q.

Proposition 1

$$D(p^*, q) \;<\; \frac{1}{1+\varepsilon}\, D(p, q).$$

Proof.

Since p is not a $(1+\varepsilon)$-ANN, by definition $D(p,q) > (1+\varepsilon)\, D(p^*, q)$. Rearranging yields the claim. □

Proposition 2 (Triangle bounds)

$$\tfrac{\varepsilon}{1+\varepsilon}\, D(p, q) \;<\; D(p, p^*) \;<\; \tfrac{2+\varepsilon}{1+\varepsilon}\, D(p, q).$$

Proof.

By the triangle inequality, $D(p, p^*) \geq \Big| D(p,q) - D(p^*, q) \Big| = D(p,q) - D(p^*, q)$. Using Proposition 1, $D(p^*, q) < \frac{1}{1+\varepsilon} D(p, q)$, so

$$D(p, p^*) \;>\; \Big(1 - \frac{1}{1+\varepsilon}\Big) D(p, q) \;=\; \frac{\varepsilon}{1+\varepsilon}\, D(p, q).$$

For the upper bound, again by the triangle inequality,
$D(p, p^*) \leq D(p,q) + D(p^*, q) < \Big(1 + \frac{1}{1+\varepsilon}\Big) D(p, q) = \frac{2+\varepsilon}{1+\varepsilon}\, D(p, q).$ □

Contraction Lemma (linking α to descent wrt q)

Lemma 6

If there exists $u \in P$ with

$$D(u, p^*) \;\le\; \tfrac{1}{\alpha}\, D(p, p^*), \quad \alpha > 1,$$

then

$$\frac{D(u,q)}{D(p,q)} \;\le\; \phi(\alpha, \varepsilon) \;:=\; \tfrac{1}{\alpha} + \tfrac{1+1/\alpha}{1+\varepsilon}.$$

Proof.

By the triangle inequality, $D(u,q) \le D(u,p^*) + D(p^*,q)$. By the premise, $D(u,p^*) \le \frac{1}{\alpha} D(p,p^*)$. Applying Proposition 2 to bound $D(p,p^*)$ from above and Proposition 1 to bound $D(p^*,q)$, we obtain

$$D(u,q) \le \tfrac{1}{\alpha} \cdot \tfrac{2+\varepsilon}{1+\varepsilon}\, D(p,q) \;+\; \tfrac{1}{1+\varepsilon}\, D(p,q).$$

Dividing both sides by $D(p,q)$ gives

$$\frac{D(u,q)}{D(p,q)} \le \frac{2+\varepsilon}{\alpha(1+\varepsilon)} + \frac{1}{1+\varepsilon} = \frac{(2+\varepsilon)/\alpha + 1}{1+\varepsilon} = \frac{(1+\varepsilon)/\alpha}{1+\varepsilon} + \frac{1+1/\alpha}{1+\varepsilon} = \frac{1}{\alpha} + \frac{1+1/\alpha}{1+\varepsilon}.$$

□

Proposition 3 (Descent threshold)

$\phi(\alpha, \varepsilon) < 1 \iff \alpha > 1 + \frac{2}{\varepsilon}$.

Proof.

Compute:

$$\phi(\alpha, \varepsilon) = \frac{1}{\alpha} + \frac{1+1/\alpha}{1+\varepsilon}.$$

Multiply both sides of $\phi(\alpha,\varepsilon) < 1$ by $1+\varepsilon > 0$:

$$\frac{1+\varepsilon}{\alpha} + 1 + \frac{1}{\alpha} \;<\; 1+\varepsilon \quad\iff\quad \frac{2+\varepsilon}{\alpha} \;<\; \varepsilon \quad\iff\quad \alpha \;>\; \frac{2+\varepsilon}{\varepsilon} \;=\; 1 + \frac{2}{\varepsilon}.$$

The reverse implication follows by reversing these equivalences. □

Analogy — Landmark Signposts Toward an Unseen Query

Story. The query q is not on our map, but its nearest landmark $p^* \in P$ is. We place one-way *signposts* (out-edges) so that from any location p you either have a direct sign to p^* or to some u that is much closer to p^* than p is (*shortcut property*).

Why this shrinks distance to q. Moving toward p^* also pulls us closer to the hidden destination q:

$$d(u,q) \;\le\; d(u,p^*) + d(p^*,q) \;\le\; \tfrac{1}{\alpha}\, d(p,p^*) + d(p^*,q) \;\le\; \phi(\alpha,\varepsilon)\, d(p,q) \;<\; d(p,q),$$

so each hop strictly reduces the distance to q until we reach a $(1+\varepsilon)$-ANN.

One-line takeaway. *You don't need q in the graph; signposts that reliably pull you toward p^* are enough to keep shrinking the distance to q.*

Math in one line

$$d(u,q) \le d(u,p^*) + d(p^*,q) \le \tfrac{1}{\alpha} d(p,p^*) + d(p^*,q) \le \phi(\alpha,\varepsilon)\, d(p,q) < d(p,q).$$

One-Hop Contraction (specialize $v = p^*$)

Lemma 7 (One-hop contraction on P)

Assume the shortcut property with parameter $\alpha > 1$. If $\alpha > 1 + \frac{2}{\varepsilon}$ and p is not an $(1+\varepsilon)$-ANN for q, then there exists $u \in N^+(p)$ with $D(u,q) < D(p,q)$.

Proof.

Apply Lemma 5 to the pair (p, v) with $v = p^* = \mathrm{NN}(q)$. There are two cases.
Case 1: $p^* \in N^+(p)$. Since p is not an $(1+\varepsilon)$-ANN, Proposition 1 gives $D(p^*, q) < \frac{1}{1+\varepsilon} D(p,q) < D(p,q)$, so $u = p^*$ is a strictly better neighbor.
Case 2: There exists $u \in N^+(p)$ with $D(u, p^*) \leq \frac{1}{\alpha} D(p, p^*)$. By Lemma 6, $D(u,q) \leq \phi(\alpha, \varepsilon)\, D(p,q)$. If $\alpha > 1 + \frac{2}{\varepsilon}$, then by Proposition 3, $\phi(\alpha, \varepsilon) < 1$, hence $D(u,q) < D(p,q)$. □

Termination and Correctness

Proposition 4

On finite P, assume the shortcut property holds with some $\alpha > 1 + \frac{2}{\varepsilon}$. Then a greedy descent (and any beam search that always expands at least the current best candidate) terminates at an $(1+\varepsilon)$-ANN for any q.

Proof.

Consider the greedy walk that starts at any p_0 and repeatedly moves to a neighbor with strictly smaller distance to q whenever possible. By Lemma 7, if the current node p is not an $(1+\varepsilon)$-ANN, there exists $u \in N^+(p)$ with $D(u,q) < D(p,q)$. Thus the sequence of visited distances is strictly decreasing and bounded below by 0. Because P is finite, there cannot be an infinite strictly decreasing sequence of the discrete set $\{D(x,q) : x \in P\}$; the process must terminate.
Suppose the walk terminates at some p_{term} that is *not* an $(1+\varepsilon)$-ANN. Then Lemma 7 provides $u \in N^+(p_{\mathrm{term}})$ with $D(u,q) < D(p_{\mathrm{term}}, q)$, contradicting termination. Hence p_{term} is an $(1+\varepsilon)$-ANN.
For beam search that always extracts the closest unvisited candidate, the greedy path is a subsequence of the expanded nodes: whenever the greedy move exists, that next node is no farther than all current candidates and will be extracted in finite steps. Therefore the same termination and correctness conclusion applies. □

Takeaways (Ordered Chain)

1. **Out-edge selection** (at each p) via α-shortcut greedy.
2. $\Rightarrow$ **Shortcut property** holds (Lemma 5).
3. $\Rightarrow$ With $v = p^*$, obtain **Contraction** (Lemma 6).
4. $\Rightarrow$ If $\alpha > 1 + \frac{2}{\varepsilon}$, per-hop **descent** (Prop. 3, Lemma 7).
5. $\Rightarrow$ **Termination &** $(1+\varepsilon)$-**ANN** correctness (Prop. 4).